

CrabTransformer Training Report

Scenario: REAL | Loss: Tweedie | Lag: 5 Years | Current Spawner: Yes

— Training Configuration —

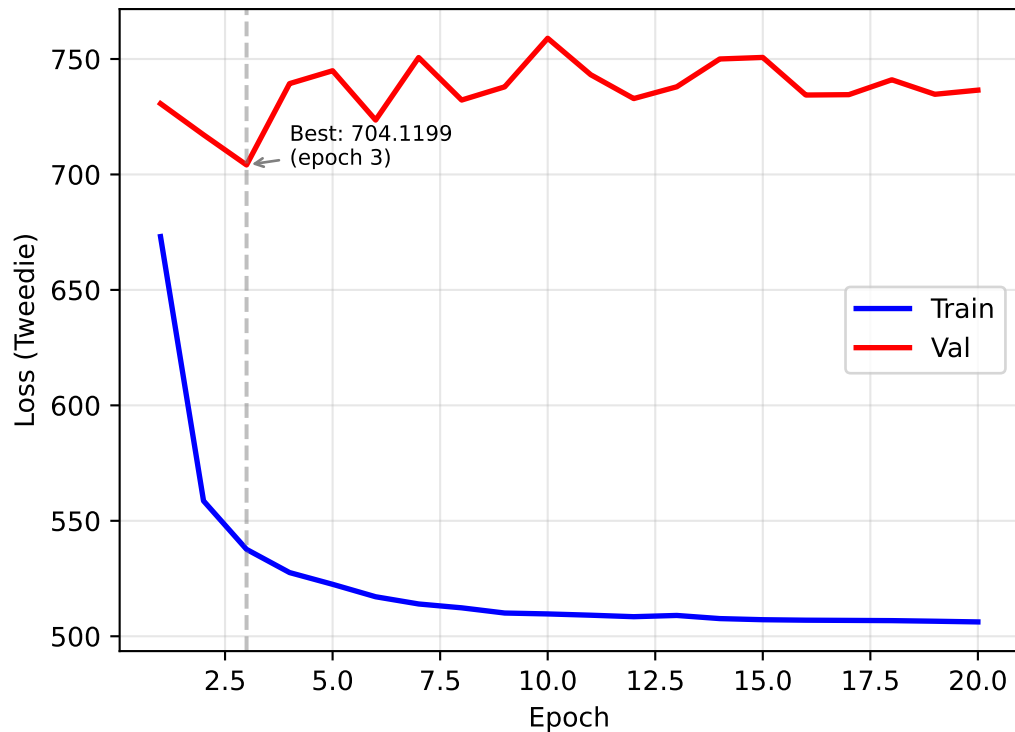
epochs_run: 20 best_val_epoch: 3
best_val_loss: 704.1199 final_train_loss: 506.2114
bias_correction (MSE): 1.0000
include_current_spawner: True

— Model Architecture —

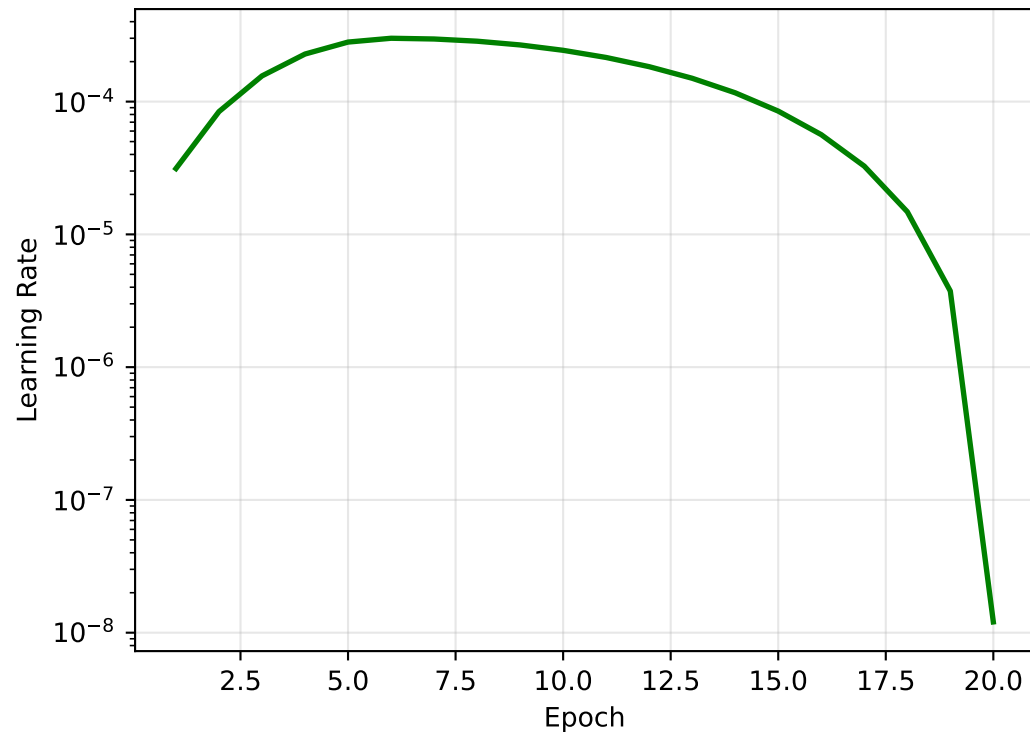
embed_dim: 128 num_heads: 4 num_layers: 3
d_ff: 512 dropout: 0.2 patch_size: 5
train/val/test years: 21/6/4

Training Curves: REAL | Lag 5 | Tweedie | w/ Spawner(t)

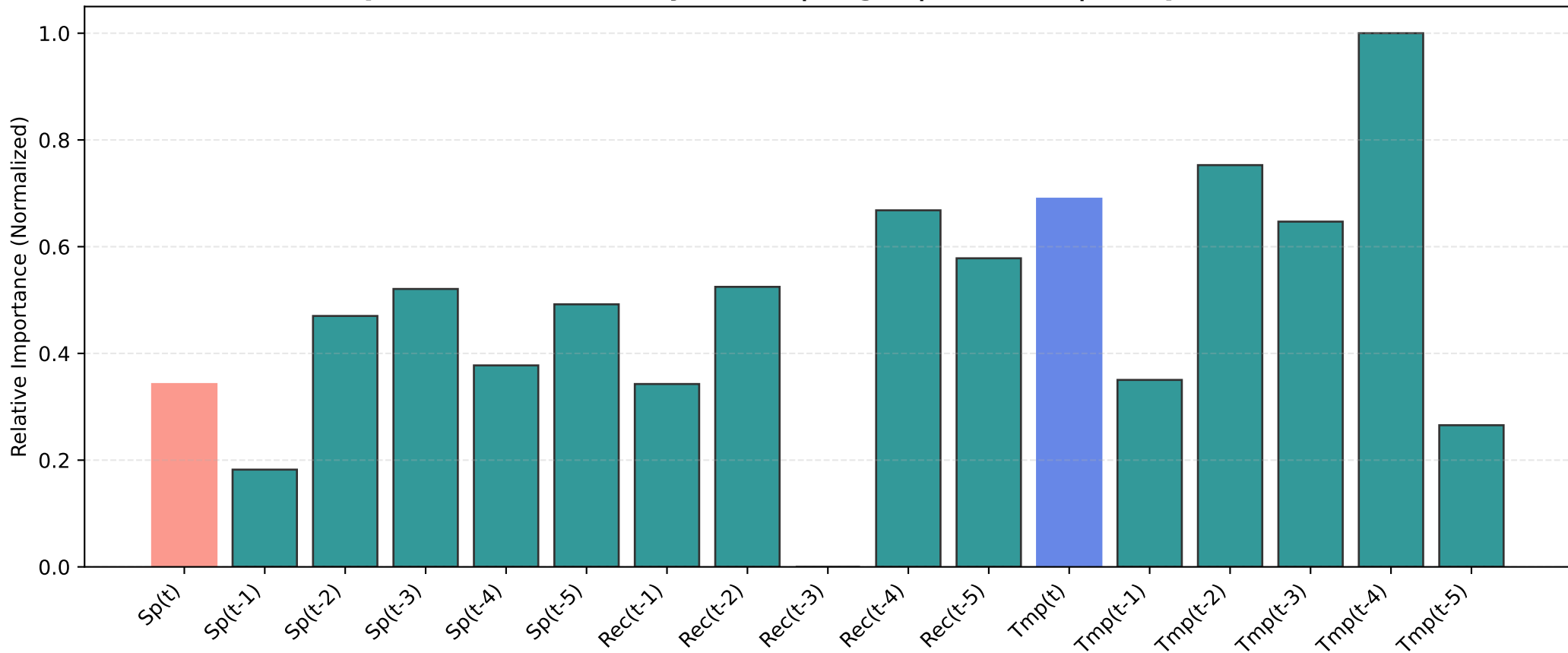
Training & Validation Loss



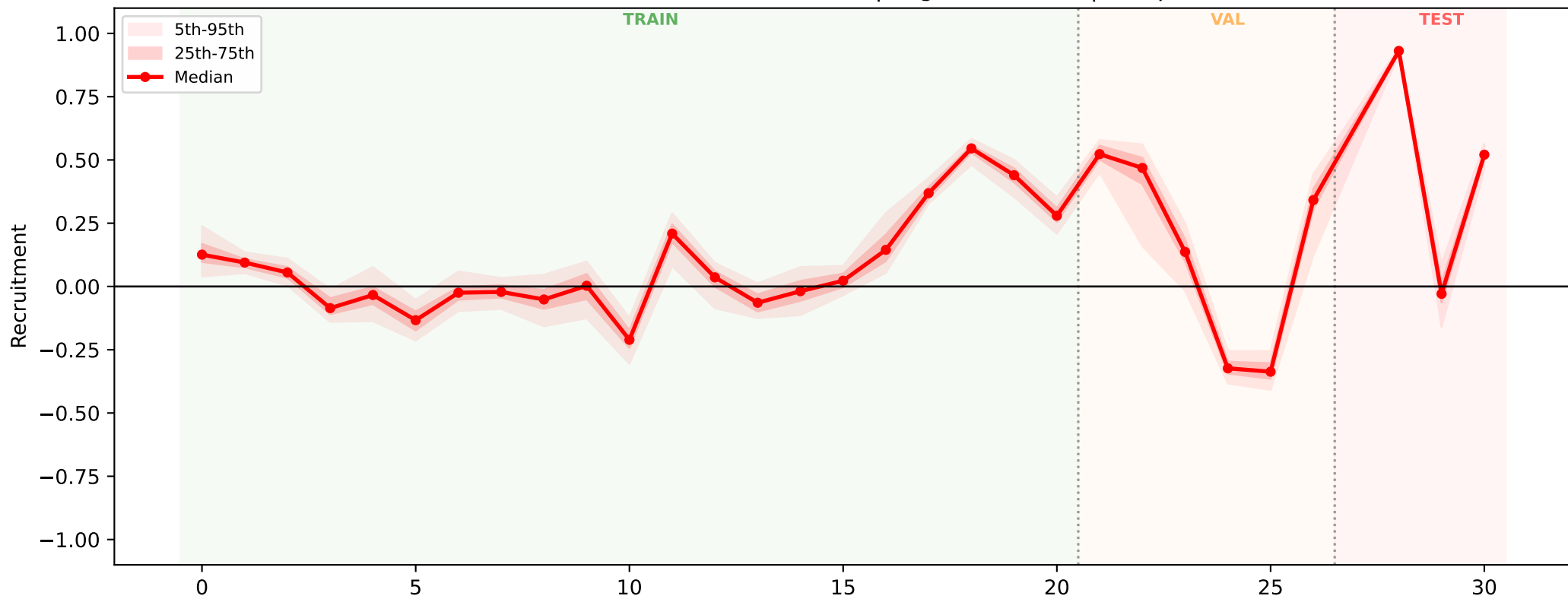
OneCycleLR Schedule



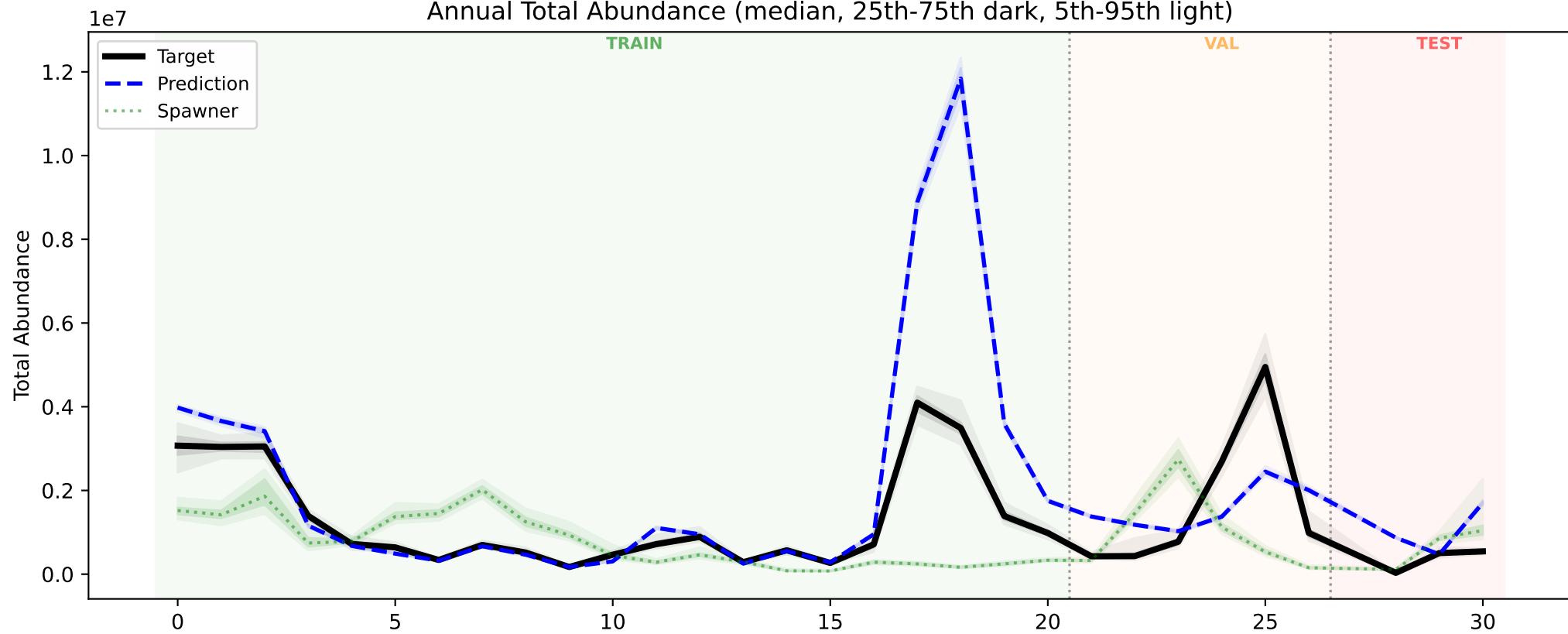
Input Channel Priority: REAL | Lag 5 | Tweedie | w/ Spawner(t)



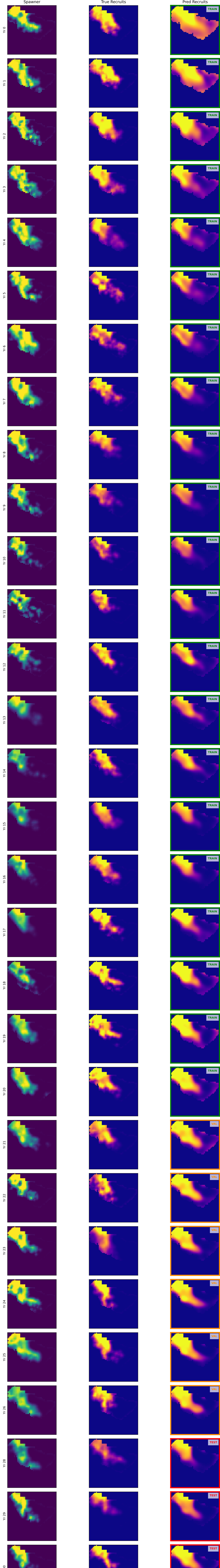
Annual Recruitment Deviations: REAL | Lag 5 (Tweedie) | w/ Spawner(t)



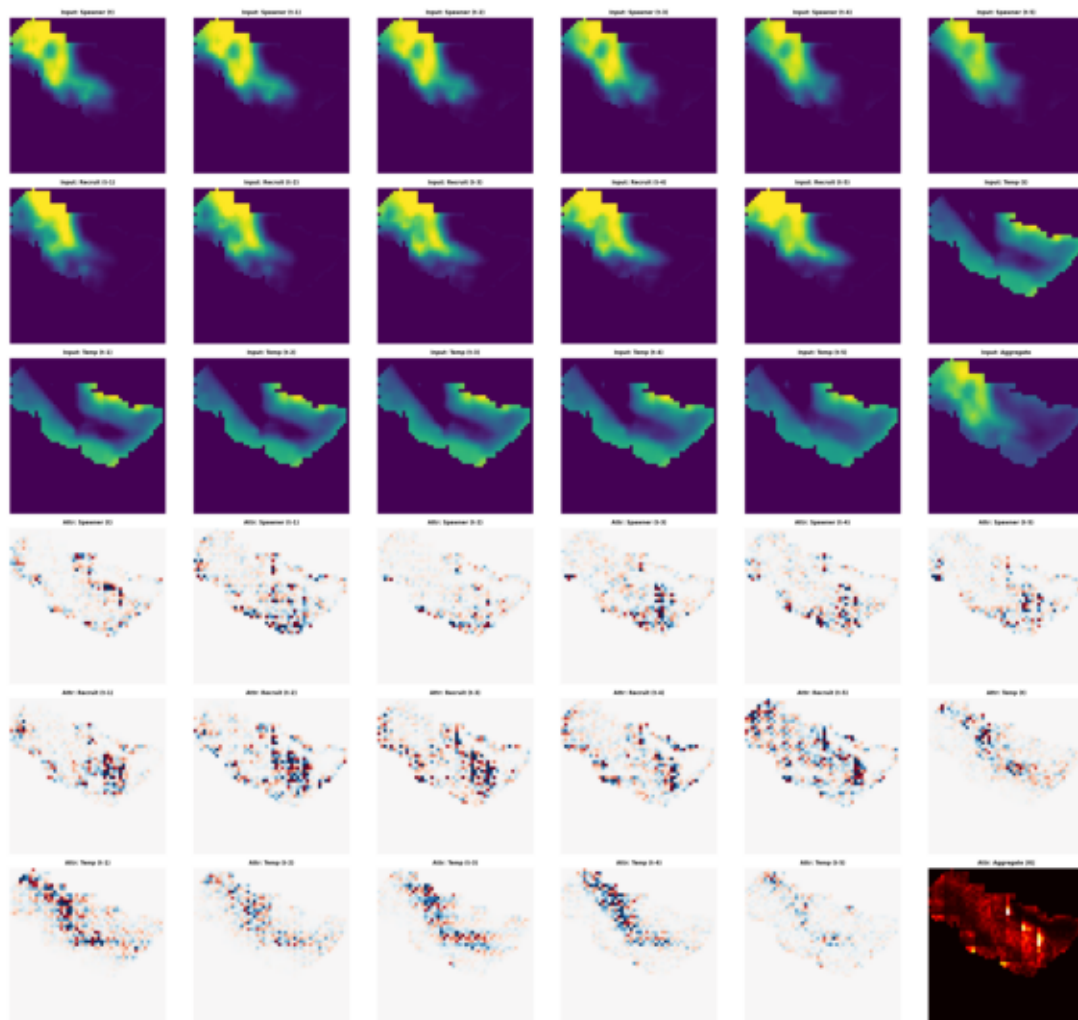
Annual Total Abundance (median, 25th-75th dark, 5th-95th light)



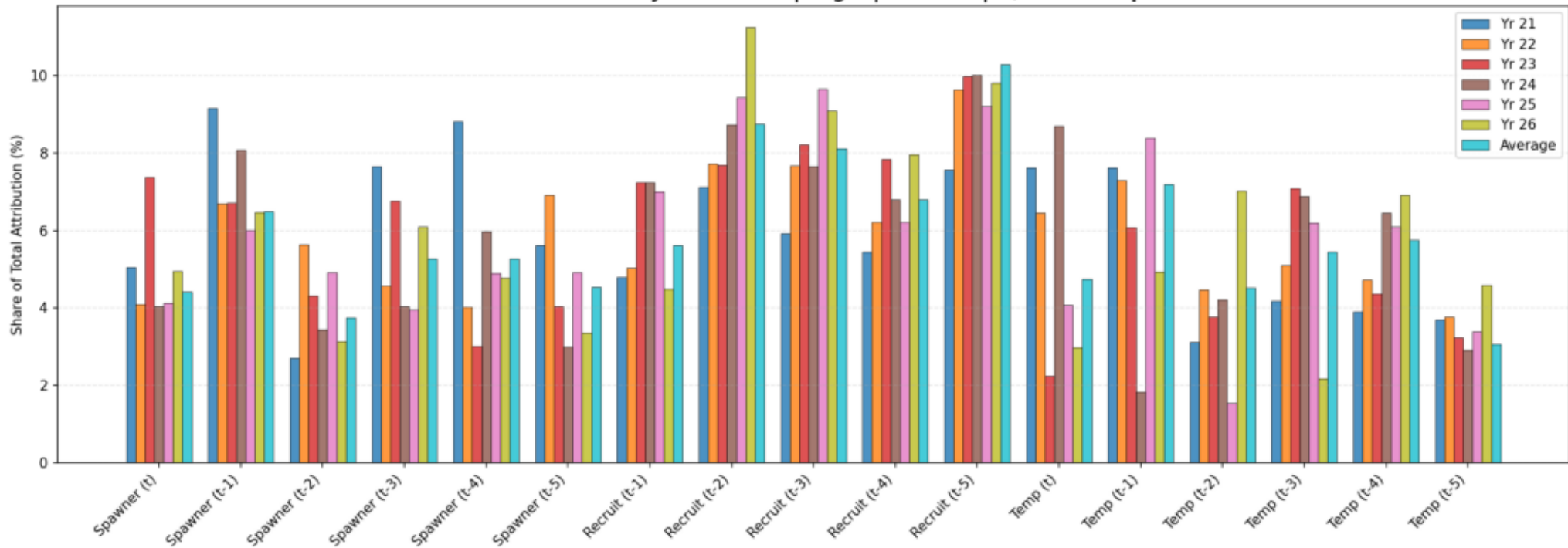
Spatial Trajectory: REAL | Lag 5 | Tweedie | w/ Spawner(t)



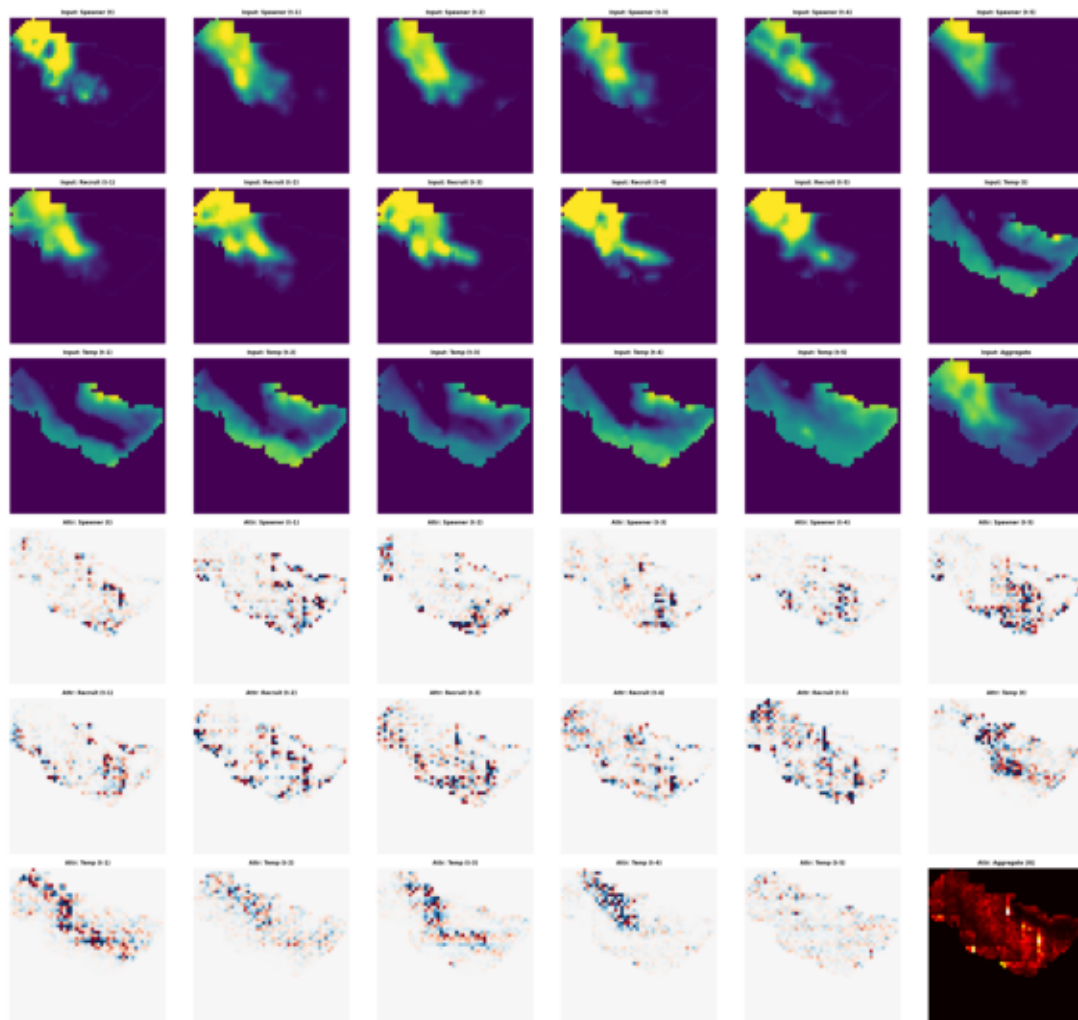
Year AVG — REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



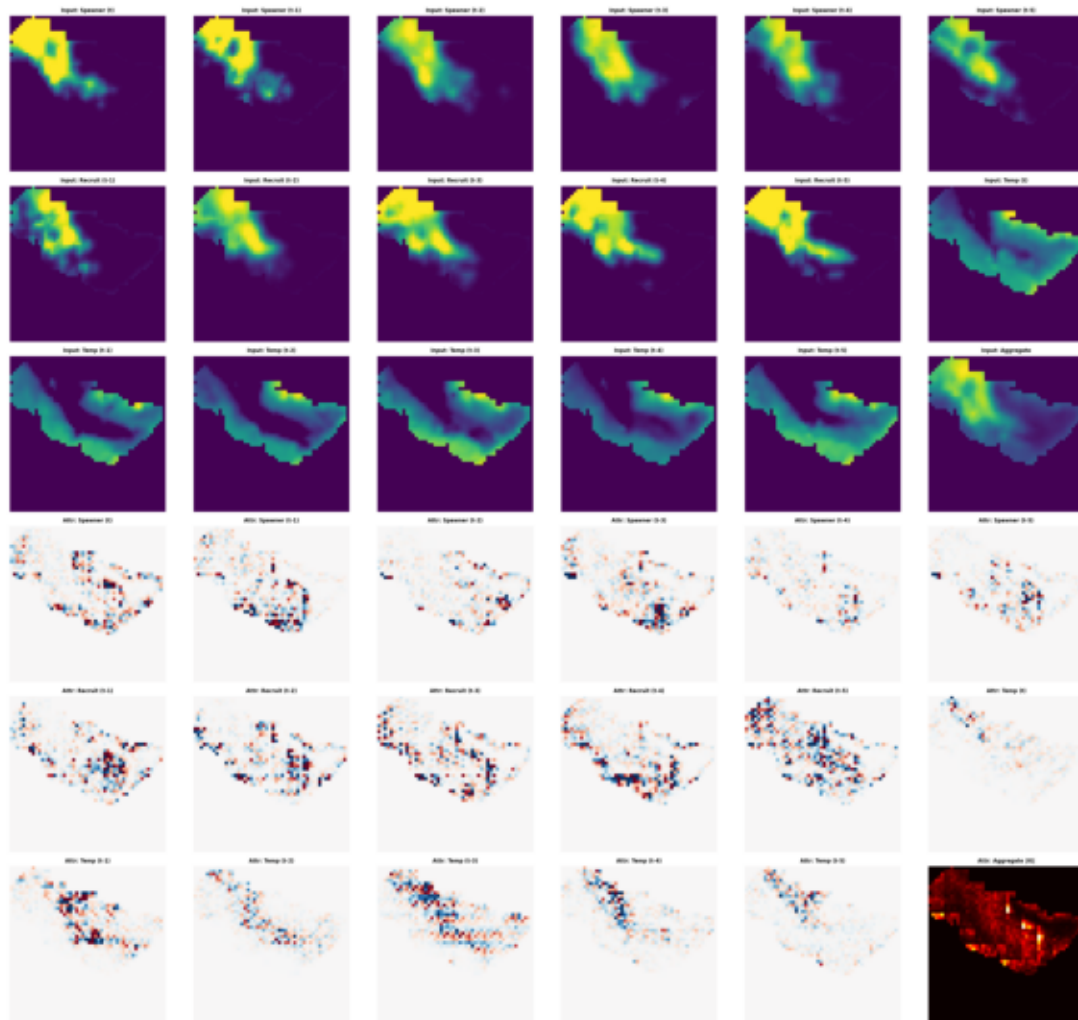
Channel Attribution by Year: REAL | Lag 5 | Tweedie | w/ Current Spawner



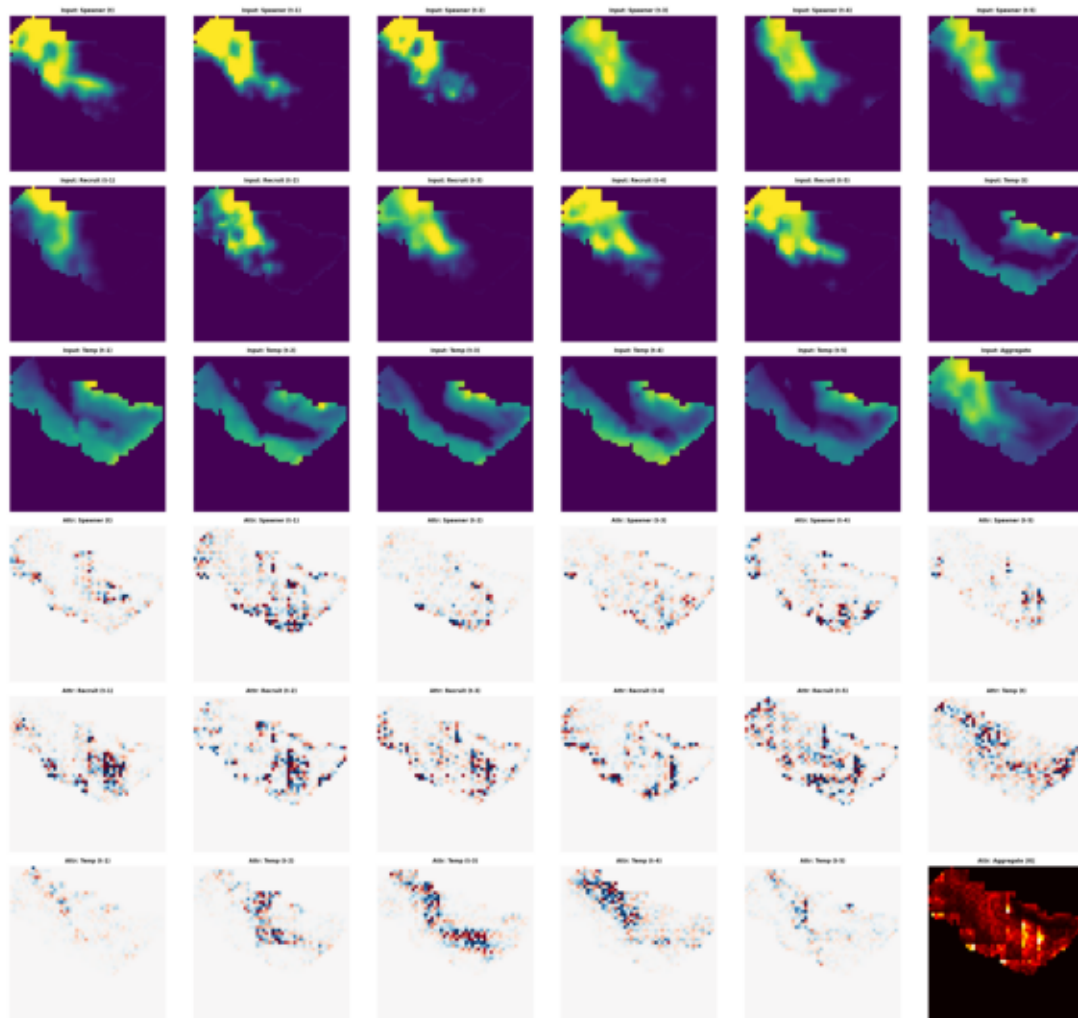
Year 22 -- REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



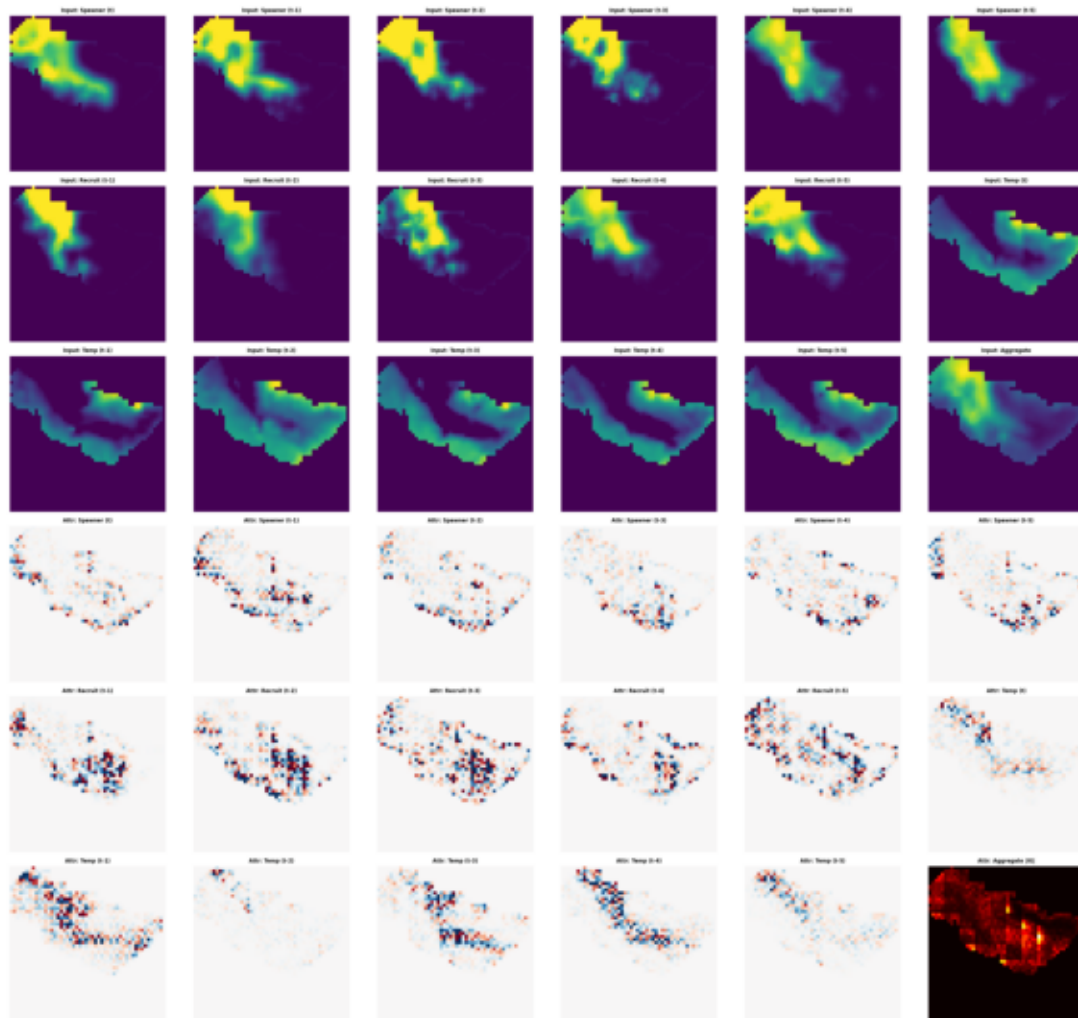
Year 23 -- REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



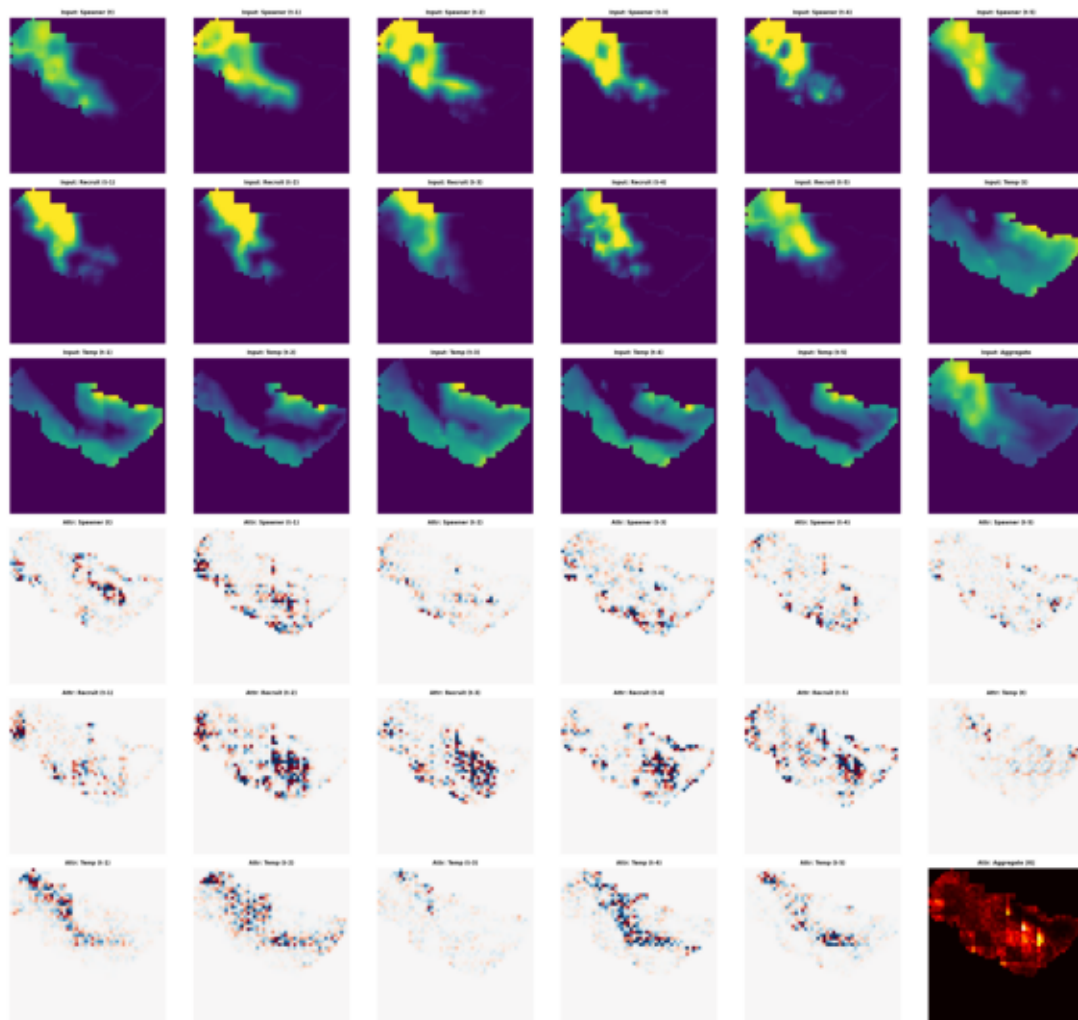
Year 24 — REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 25 -- REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Year 26 -- REAL | Lag 5 | Tweedie | w/ Current Spawner
 Top: Mean Input | Bottom: Attribution (red=+recruit, blue=-recruit)



Summary Statistics: REAL | Lag 5 | Tweedie | w/ Spawner(t)

Phase	Pred_Mean	Tgt_Mean	Pred_SD	Pred_P50	Pred_P95	Abundance_Capture	MAE	Spearman
TRAIN	2440.2	1480.3	12016.1	0.0	11846.4	1.648	1363.6	0.898
VAL	1771.8	1950.8	4518.1	0.0	9814.7	0.908	2098.0	0.789
TEST	1148.3	406.3	3659.1	0.0	7423.1	2.826	928.0	0.805

Detailed Metrics (R-style): REAL | Lag 5 | Tweedie | w/ Spawner(t)

Phase	RMSE	RMSE_unbias	Bias	Bias_%	Spearman	Pearson	Pct_Error	Zero_F1	TopK_Acc	TopK_Jaccard	Quant_Err	Skill_MSE	Skill_MAE
TRAIN	4027.7	3871.7	959.9	3.7%	0.904	0.906	29.6%	0.868	0.812	0.691	0.70	0.151	0.577
VAL	6217.8	6077.6	-179.0	5.8%	0.793	0.652	65.2%	0.838	0.589	0.421	1.24	-0.981	0.080
TEST	3126.4	3002.1	742.0	6.7%	0.812	0.636	1001.5%	0.881	0.693	0.554	0.99	-407.891	-5.630